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DeepH-pack: A general-purpose neural network package for deep-learning electronic structure calculations

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Abstract

In computational physics and materials science, first-principles methods, particularly density functional theory, have become central tools for electronic structure prediction and materials design. Recently, rapid advances in artificial intelligence (AI) have begun to reshape the research landscape, giving rise to the emerging field of deep-learning electronic structure calculations. Despite numerous pioneering studies, the field remains in its early stages; existing software implementations are often fragmented, lacking unified frameworks and standardized interfaces required for broad community adoption. Here we present DeepH-pack, a comprehensive and unified software package that integrates first-principles calculations with deep learning. By incorporating fundamental physical principles into neural-network design, such as the nearsightedness principle and the equivariance principle, DeepH-pack achieves robust cross-scale and cross-material generalizability. This allows models trained on small-scale structures to generalize to large-scale and previously unseen materials. The toolkit preserves first-principles accuracy while accelerating electronic structure calculations by several orders of magnitude, establishing an efficient and intelligent computational paradigm for large-scale materials simulation, high-throughput materials database construction, and AI-driven materials discovery.

Introduction

Density functional theory (DFT) has established itself as a standard method for first-principles calculations, playing a pivotal role in understanding complex quantum materials and designing novel devices [1, 2]. Despite its success, the computational cost of conventional DFT scales cubically with the number of electrons, becoming prohibitive for large systems. This computational complexity fundamentally restricts the applicability of DFT in demanding tasks, such as large-scale molecular dynamics, electronic-structure simulations, and high-throughput

screening across massive materials databases [3]. Consequently, the trade-off between accuracy and efficiency poses a persistent bottleneck for realistic device modeling and materials discovery. In response to this challenge, artificial intelligence (AI), particularly deep learning, has emerged as a transformative paradigm. By training on high-quality data generated by first-principles calculations, deep neural networks can capture the underlying quantum-mechanical mappings from atomic structures to physical properties, offering the potential to achieve prediction accuracy comparable to reference first-principles methods with significantly reduced computational cost.

The remarkable success of deep learning across diverse scientific domains has accelerated its adoption in *ab initio* materials simulation [4–9]. While earlier efforts focused primarily on developing machine learning interatomic potentials (MLIPs) to predict energy and forces [10–13], the frontier has recently expanded to direct prediction of electronic structure quantities [14, 15]. There is a growing focus on using neural networks to reproduce high-dimensional, multi-faceted properties with quantum-mechanical origins, including charge densities [16, 17], density matrices [18, 19], wave functions [20, 21], and band structures [22]. These developments allow machine-learning-assisted simulations to go beyond thermodynamic and mechanical properties, enabling the investigation of a wider range of electronic, magnetic, and optical phenomena.

Among these electronic quantities, the DFT Hamiltonian holds a central position. As the fundamental operator governing quantum material systems, the DFT Hamiltonian encodes essential information about the electronic structure properties. Once the Hamiltonian is determined, in principle all physical observables at the DFT level, such as band structures, response functions, and transport properties, can be derived through standard post-processing calculations [3, 23]. Motivated by this capability, the deep-learning DFT Hamiltonian (DeepH) method pioneered the direct prediction of DFT Hamiltonians via deep learning [9]. As the first approach to generalize from small-scale material systems to large ones and achieve strict quantum-mechanical fidelity in this domain, DeepH exemplifies the *Physics for AI* paradigm: It leverages fundamental physical principles, including the nearsightedness principle of electronic matter and the equivariance principle, to guide the design of neural network architecture and the selection of target quantities for deep learning. This physics-informed design ensures high data efficiency and extraordinary generalizability of the DeepH approach, as demonstrated in the original work, which provided the first open-source code addressing this problem [9]. At present, several related algorithms have been developed, including the series of DeepH algorithms [24, 25], QHNet [26], HamGNN [27, 28], Hot-Ham [29], among others [30]. By successfully embedding physical priors into deep-learning models, DeepH establishes a virtuous cycle, employing AI as a reliable tool to accelerate and expand the frontiers of physical research, leading to various applications in electronic structure analysis [31–35].

Building on this methodological foundation, DeepH has evolved into a general-purpose algorithmic framework with high versatility and scalability. It is capable of modeling systems across various length scales, ranging from small molecules to complex solids containing tens of thousands of atoms, while handling intricate physical effects such as spin-orbit coupling (SOC) and magnetic ordering. Furthermore, the framework’s applicability extends well beyond the sole prediction of the DFT Hamiltonian. When paired with corresponding data flows, the DeepH method can be generalized to predict a wide array of physical quantities that transform as scalars or tensors, including

energies and atomic forces [36], density matrices [19], real-space Kohn-Sham potentials [37], charge densities, as well as physical quantities within the generalized Kohn-Sham framework [38] and density functional perturbation theory [39]. Operationally, it allows for seamless integration with existing DFT software ecosystems, where the predicted quantities can serve as high-quality initial guesses to accelerate self-consistent field (SCF) convergence or be used directly for post-processing tasks. This establishes a symbiotic workflow that combines the efficiency of deep learning with the rigorous formalism of first-principles calculations.

While significant progress has been made at the algorithmic level [9, 24, 25, 36, 38–41], realizing AI-driven materials discovery still requires substantial foundational engineering efforts in software and implementation. Bridging the gap between methodological advances and practical applications demands more than theoretical and algorithmic sophistication: it critically depends on the availability of high-performance, scalable, and user-friendly software implementations. Although promising approaches for predicting key electronic properties continue to emerge, the development of a unified deep learning framework, characterized by versatility, scalability, and ease of use across diverse material systems and length scales, would greatly benefit the research community. Such a framework has the potential to significantly accelerate the adoption of these advanced methods in practical *ab initio* simulations.

In this article, we present DeepH-pack, a comprehensive and production-ready software package designed to predict DFT Hamiltonians for atomistic systems by leveraging advanced graph neural network (GNN) architectures and symmetry-aware methods. The fundamental advance of DeepH-pack lies in its transition of the deep-learning electronic structure method from an academic proof-of-concept demonstration into a production-ready, high-performance scientific computing infrastructure. While prior algorithms successfully established the theoretical validity of mapping atomic configurations to quantum-mechanical Hamiltonians, DeepH-pack systematically addresses the core engineering and algorithmic bottlenecks in broad community adoption.

Specifically, the software infrastructure of DeepH-pack introduces four key innovations over previous independent implementations: (i) *High-performance re-engineering* via a full functional refactoring under the JAX framework and XLA compilation to accelerate training while ensuring enhanced numerical stability; (ii) *Channel-flexible equivariant convolutions* through an optimized tensor product module that curtails 60%–75% of redundant high-order representation parameters by embedding physical hierarchy; (iii) *Network customization* offering light-weighted **Normal** and high-performance **Accurate** architectures to balance expressivity and memory footprint; and (iv) *End-to-end workflow integration* by establishing a dedicated companion toolkit, DeepH-dock, that provides seamless bidirectional data interfaces with multiple mainstream DFT suites alongside post-processing modules to directly derive physical properties from predicted Hamiltonians.

Ultimately, these concerted developments bridge advanced machine learning with daily research demands, delivering a streamlined and highly intuitive computational experience in practical materials research applications. By exploiting the localized nature of the underlying electronic structures, the time complexity of the implementation grows strictly linearly with the number of atoms, greatly improving upon the cubic complexity that is typical of traditional DFT methods. Empirically, even for small systems containing only tens of atoms, the package accelerates Hamiltonian predictions by two to three orders of magnitude compared to standard DFT. Despite its substantial speedup, DeepH-pack maintains high accuracy, achieving sub-meV agreement with conventional DFT methods in electronic band structures.

To seamlessly extend these computational advantages to a wide research ecosystem, DeepH-pack provides automated interfaces for major open-source DFT software suites, including FHI-aims [42, 43], Quantum ESPRESSO [44], SIESTA [45], ABACUS [46], and OpenMX [47]. This compatibility allows users to integrate DeepH-pack easily into existing computational workflows. By supporting both crystalline solids and molecular systems, DeepH-pack serves as a versatile and powerful tool for the computational materials science community. The computational workflow of DeepH-pack is schematically displayed in Fig. 1.

Results

The DeepH framework

A brief outline of the theoretical background of the DeepH method is provided here. The main capability of DeepH-pack is to predict the DFT Hamiltonian $\hat{H}_{\text{DFT}}$, a core quantity in the Kohn-Sham formalism [1]. In this theory, $\hat{H}_{\text{DFT}}$ is an effective one-electron operator that serves as a compact gateway to many properties of interest at the DFT level. From its eigenvalues $\{\mathcal{E}_n\}$ and eigenstates $\{|\psi_n\rangle\}$, one can compute the electron density, total energy, Hellmann-Feynman forces, and, for periodic crystals, the band structure and the density of states (DOS). Practically, $\hat{H}_{\text{DFT}}$ is a cornerstone for deriving many properties of interest, including charge and spin densities, forces and stresses, and a variety of transport and response functions [48–50]. Therefore, DeepH-pack focuses on predicting the DFT Hamiltonian as a fundamental step toward enabling accurate and efficient electronic structure modeling (Fig. 2(a)).

In conventional DFT calculations, to obtain the Hamiltonian, one needs to solve the Kohn-Sham equation

$$\hat{H}_{\text{DFT}}|\psi_n\rangle = \mathcal{E}_n|\psi_n\rangle \quad (1)$$

self-consistently, which is time-consuming due to its cubic complexity in the number of atoms and the cumbersome self-consistent iterations. However, a more straightforward route exists with the help of AI. According to the Hohenberg-Kohn theorem, $\hat{H}_{\text{DFT}}$ is uniquely determined by the material structure, i.e., the atomic species $\{Z_i\}$

and the atomic positions $\{\mathcal{R}_i\}$ [1]. Therefore, DeepH-pack aims to model a direct mapping $\{Z_i, \mathcal{R}_i\} \mapsto \hat{H}_{\text{DFT}}(\{Z_i, \mathcal{R}_i\})$ using deep learning. To make full use of the locality principle, DeepH-pack follows the original work of Li *et al.* [9] and predicts the matrix elements. Given atoms i and j located at positions $\mathcal{R}_i$ and $\mathcal{R}_j$ within their respective unit cells, which are displaced by lattice vectors $\mathbf{R}_i$ and $\mathbf{R}_j$, the Hamiltonian matrix elements are defined as:

$$H_{i\alpha,j\beta}(\mathbf{R}_j - \mathbf{R}_i) = \int d^3\mathbf{r} d^3\mathbf{r}' \phi_{i\alpha}^*(\mathbf{r} - \mathcal{R}_i) \hat{H}_{\text{DFT}} \phi_{j\beta}(\mathbf{r}' - \mathcal{R}_j) \quad (2)$$

under a basis of localized atomic-orbital functions $\{\phi_{i\alpha}\}$. Here, α, β label orbitals on those atoms. When SOC is present, α and β have an additional spin degree of freedom. In the remainder of this paper, we use H_{ij} to denote the sub-blocks of Hamiltonian matrix elements associated with the atom-pair i and j .

After obtaining the DFT Hamiltonian, one can solve the non-self-consistent Kohn-Sham equation (1) to obtain its eigenspectrum for further study. For periodic systems, Eq. (1) takes the form of a generalized eigenvalue problem [51]:

$$\sum_{j\beta \in \text{uc}} H_{i\alpha,j\beta}(\mathbf{k}) c_{n,j\beta}(\mathbf{k}) = \mathcal{E}_n(\mathbf{k}) \sum_{j\beta \in \text{uc}} S_{i\alpha,j\beta}(\mathbf{k}) c_{n,j\beta}(\mathbf{k}), \quad (3)$$

$\mathcal{E}_n(\mathbf{k})$ are called the electronic band structure, and $c_{n,j\beta}(\mathbf{k})$ are the wave function coefficients in the given basis. Here, “uc” denotes the unit cell at the origin, and $i\alpha \in \text{uc}$. The matrix elements in reciprocal space are constructed from the real-space Hamiltonian and overlap matrices via the Fourier transform:

$$O_{i\alpha,j\beta}(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k} \cdot \mathbf{R}} O_{i\alpha,j\beta}(\mathbf{R}), \quad O \in \{H, S\}, \quad (4)$$

where $\mathbf{R}$ denotes the lattice vector connecting the unit cells. The real-space overlap matrix elements $S_{i\alpha,j\beta}(\mathbf{R})$ are defined as:

$$S_{i\alpha,j\beta}(\mathbf{R}_j - \mathbf{R}_i) = \int d^3\mathbf{r} \phi_{i\alpha}^*(\mathbf{r} - \mathcal{R}_i) \phi_{j\beta}(\mathbf{r} - \mathcal{R}_j), \quad (5)$$

where $\phi_{i\alpha}$ represents the α -th orbital of the i -th atom located at $\mathcal{R}_i$.

For non-periodic systems, Eq. (3) naturally reduces to the case where the unit cell is sufficiently large that no orbital overlaps with periodic images, and the $\mathbf{k}$ -dependence vanishes.

We build our method on equivariant graph neural networks. To enhance the usability, predictive capacity, and scalability of deep-learning electronic structure calculations, DeepH-pack incorporates a series of innovations in neural network algorithms and software design. In the following paragraphs, we outline these core advancements, detailing the fundamental enforcement of physical symmetries, optimizations for high-order tensor products, and the deployment of customized model architectures tailored to diverse computational demands.

DeepH-pack predicts DFT Hamiltonians by leveraging a message-passing GNN where the atomic structure is mapped to graph data. In this representation, nodes correspond to atoms and edges connect atoms with finite Hamiltonian matrix elements. This GNN architecture naturally reflects the locality of the Hamiltonian and is consistent with the nearsightedness principle of electronic matter [52]. Through multiple layers of message-passing operations, the information from neighboring atoms is aggregated to update the node and edge features. This process effectively captures the complex local environment of the Hamiltonian matrix sub-blocks H_{ij} , allowing the model to predict accurate interactions even for atoms that are spatially separated but quantum-mechanically coupled.

A fundamental requirement for predicting physical quantities is respecting the symmetry of the system under global Euclidean transformations in 3-dimensional space (the $E(3)$ group). The Hamiltonian sub-block H_{ij} is translationally invariant and rotationally equivariant [9, 13]. To guarantee translational invariance, the network inputs rely solely on relative coordinate vectors $\mathcal{R}_{ij} = \mathcal{R}_j - \mathcal{R}_i$ avoiding absolute positions entirely. Regarding rotation, if the atomic coordinates are rotated by a rotation matrix $R \in \text{SO}(3)$, where $\text{SO}(3)$ denotes the proper rotation group in 3-dimensional space, the Hamiltonian transforms according to

$$H_{ij}(\{R \cdot \mathcal{R}_k\}) = D_i(R)H_{ij}(\{\mathcal{R}_k\})D_j^\dagger(R), \quad (6)$$

where $D_i(R)$ and $D_j(R)$ are the representation matrices associated with the atomic-orbital basis sets centered at atoms i and j , respectively. These matrices are constructed as direct sums of Wigner D matrices [9, 53]. To satisfy this constraint, DeepH-pack decomposes the target Hamiltonian into vectors transforming under irreducible representations (irreps) of the $\text{SO}(3)$ group [24]. For systems involving SOC, the spin degrees of freedom introduce half-integer representations, but they can be transformed back into integer representations that fit into the framework of equivariant neural networks (e.g., $1/2 \otimes 1/2 = 0 \oplus 1$) [24], ensuring that the predicted quantities strictly obey the physical transformation laws.

To achieve the symmetry constraints imposed on the outputs, DeepH-pack utilizes the equivariant neural network (ENN) architecture [54], wherein the internal features are constructed as $\text{SO}(3)$ -equivariant vectors. These features carry specific irreps of the $\text{SO}(3)$ group and transform according to Wigner D -matrices.

Given a node or an edge, the feature x_i or e_{ij} (uniformly denoted as a here) is a three dimensional array. This array is indexed by a channel number c ($1 \leq c \leq C$, where C is the number of neural network channels), an angular quantum number l ($0 \leq l \leq L$, where L is a manually selected natural number), and a magnetic quantum number m ($-l \leq m \leq l$). We denote an individual element as $a_{m,c}^l$. For a given channel c and angular momentum l , the one-dimensional array $\{a_{m,c}^l\}_{m=-l,\dots,l}$ constitutes a rank- l irreducible tensor of the $\text{SO}(3)$ group, carrying the

corresponding irrep:

$$(R \cdot a)_{m,c}^l = \sum_{m'=-l}^l D_{mm'}^l(R) a_{m',c}^l \quad (7)$$

Here, $R \cdot a$ represents the value of the variable in the network after applying a spatial rotation $R \in \text{SO}(3)$ to the input atomic structure. The ENN outputs a series of irreducible tensors, including node and edge feature outputs. The high-dimensional equivariant features are then mapped back to the physical basis of the Hamiltonian via the inverse Clebsch-Gordan decomposition [24], completing the prediction process.

Architectural advancements

The first major advancement presented in this framework is the channel-flexible eSCN convolution algorithm. Linear transformations are fundamental to neural networks for mixing feature information and enhancing representational capacity. However, according to Schur's lemma [53], linearly combining irreducible representations of different orders violates equivariance. This mathematical restriction limits the expressive power of simple linear layers. To enable interactions between different angular momentum channels, we employ tensor products as the primary update mechanism. In DeepH-pack, we utilize tensor products between edge features and spherical harmonics to modify the order of the irreducible representations.

Within the DeepH architecture, equivariant convolutions operate strictly on edge features, transforming an input with C_{in} channels into an output with C_{out} channels. For a target angular momentum l where $l = 0, 1, \dots, L$, the operation is defined as:

$$\begin{aligned} \text{E3Conv}(a_{ij}; w)_{m,c}^l &= \left[\sum_{l_1 l_2} \sum_{c'} w_{cc'}^{l_1 l_2 l} (a_{ij})_{c'}^{l_1} \otimes Y^{l_2}(\hat{r}_{ij}) \right]_m^l \\ &:= \sum_{l_1, l_2} \left[\sum_{\substack{m_1=-l_1 \\ m_2=-l_2}}^{l_1, l_2} C_{l_1 m_1, l_2 m_2}^{lm} \sum_{c'=1}^{C_{in}} w_{cc'}^{l_1 l_2 l} (a_{ij})_{m_1, c'}^{l_1} Y_{m_2}^{l_2}(\hat{r}_{ij}) \right] \end{aligned} \quad (8)$$

Here, $C_{l_1 m_1, l_2 m_2}^{lm}$ represents the Clebsch-Gordan coefficients, which are non zero only when $|l_1 - l_2| \leq l \leq l_1 + l_2$. The summation over l_1 and l_2 is mathematically restricted to this domain.

Evaluating this equation directly entails computational complexity scaling as $O(CL^6 + C^2L^3)$ for both time and memory. Such steep scaling severely restricts the maximum polynomial degree L and the overall network capacity. This bottleneck is particularly detrimental for predicting DFT Hamiltonians compared to predicting interatomic potentials, because matrix elements between high angular momentum orbitals demand high order tensor components. For instance, computing interactions between two d orbitals entails components up to $l = 4$ without spin-orbit coupling, and up to $l = 5$ when spin-orbit coupling is included.

To mitigate this bottleneck, the eSCN algorithm [55] re-parameterizes the weights w to map $SO(3)$ equivariant tensors into an $SO(2)$ basis. This simplifies the convolution, reducing time and space complexities to $O(CL^3)$ and $O(C^2L^3)$, respectively, assuming a uniform channel size C . Despite this favorable scaling, standard eSCN rigidly requires identical channel counts across all representation orders. In atomic-orbital DFT Hamiltonians, high angular momentum components encode crucial physical interactions, yet their corresponding representations are significantly less abundant than those of low angular momentum counterparts. Consequently, networks do not require massive high order features and operate most efficiently using a hierarchical channel distribution. For example, DeepH-E3 allocates 64 channels to $l = 0$ but only 4 to $l = 5$. Forcing a uniform width of 64 channels artificially inflates a single tensor product block from roughly 2.3×10^4 to 3.7×10^5 parameters. This artificial inflation expands the size of convolution layers by over an order of magnitude without yielding commensurate gains in representation power.

We formulate a channel-flexible eSCN module within DeepH-pack to overcome this limitation. Our implementation generalizes the eSCN mapping to support independent channel dimensions C_{in}^l and C_{out}^l for each order of angular momentum l . This structural flexibility accommodates the hierarchical channel distributions aforementioned, drastically curtailing redundant parameters. As corroborated by our benchmark evaluations (Tab. 2), this architectural innovation plays a pivotal role in the scalability of our framework. In our ablation study comparing model architectures, the channel-flexible implementation utilized in the **Accurate** model reduces the total parameter count by approximately 70% relative to the uniform-channel counterpart. Concurrently, the **Accurate** model achieves equivalent predictive accuracy while significantly lowering the inference memory footprint per atom, confirming the objective advantage of our algorithmic optimization.

The second key advancement lies in the implementation of customized model architectures. DeepH-pack provides two distinct model architectures to balance accuracy and computational cost: **Normal** and **Accurate**. The **Normal** model (as shown in Fig. S1a) is a lightweight architecture based on improved DeepH-E3 [24], with optimizations for high inference speed and accuracy on standard tasks. Specifically, these optimizations stem from a complete structural refactoring of the original DeepH-E3 implementation. By rewriting all neural network operators using JAX [56], we preserved the original architectural framework while achieving remarkable improvements in both efficiency and accuracy. As demonstrated in our benchmark test (Tab. 1), compared to the original DeepH-E3, the **Normal** network accelerates the training speed by a factor of roughly 4 to 12 and inference speed by a factor of 3 to 6 across various material systems. Furthermore, benefiting from the inherent numerical stability of the JAX framework and the pre-computation of Clebsch-Gordan coefficients in strict double precision (float64), the

numerical accuracy of equivariant tensor operations is significantly enhanced. Consequently, it consistently reduces prediction errors, bringing down the Hamiltonian matrix error by more than 50% in most tested systems (Tab. 1).

For scenarios requiring higher precision or larger basis sets, the **Accurate** model (as shown in Fig. S1b) offers enhanced expressivity. While the computational complexity of standard equivariant convolutions in the **Normal** model typically scales as $O(L^6)$ with respect to the maximum order of angular momentum L , the **Accurate** model effectively overcomes this bottleneck. Built upon an improved DeepH-2 architecture [40], it represents a significant structural evolution. A major limitation of the original DeepH-2 framework was the rigid constraint that the number of channels for all angular momenta l must remain identical. To address this, as previously described, we have completely rewritten the implementations of $SO(3)$ and $SO(2)$ representations to seamlessly integrate a channel-flexible eSCN module. This integration effectively reduces the computational complexity of convolutions to $O(L^3)$ and permits flexible channel configurations, thereby substantially improving overall parameter efficiency. By deeply optimizing the underlying tensor execution flows and introducing streamlined $SO(2)$ activation methods, these targeted software engineering efforts ensure that the architecture trains efficiently while strictly maintaining high-precision equivariance.

In practical applications, these distinct structural designs clearly delineate the application boundaries of the two architectures. The streamlined **Normal** model is particularly well-suited for the rapid training of small-scale or highly specialized systems. For tasks involving isolated molecular systems or structural perturbations of fixed compounds, the **Normal** network achieves ideal predictive performance with exceptional time efficiency and a compact parameter size. Owing to its relatively straightforward structural design, it remains highly accessible, serving as an excellent starting point for researchers and developers new to the architecture.

However, a critical limitation of the **Normal** model is its constrained representational capacity. When tasked with learning highly diverse chemical environments, such as the carbon allotropes dataset (Tab. 1), the **Normal** model is forced to drastically expand its parameter space to compensate for its limited expressivity. Even so, it still falls short of the predictive accuracy achieved by the **Accurate** model, while concurrently suffering from significantly higher memory consumption during inference. In contrast, the highly scalable **Accurate** architecture enables training on large materials datasets encompassing, for instance, tens to hundreds of thousands of atomic structures. It serves as an algorithmic and engineering foundation for developing universal electronic structure models for materials. Ultimately, the complementary nature of the **Normal** and **Accurate** architectures provides users with the flexibility to balance computational cost and desired accuracy according to their specific research tasks.

The architectural innovations described above establish DeepH as a general-purpose, efficient, and high-precision algorithmic framework (Fig. 2(b)). A key advantage is that the learning process is agnostic to the specific exchange-correlation functional employed in the first-principles calculations. Consequently, the model is capable of predicting DFT Hamiltonians derived from a wide spectrum of exchange-correlation functionals, provided that the training dataset maintains consistency. Furthermore, the applicability of DeepH extends beyond the typical DFT Hamiltonian. By adapting the output representations to match the specific tensor ranks, the framework can predict any physical quantity that transforms as an $E(3)$ -equivariant scalar or tensor. This versatility covers a diverse array of electronic structure properties, including electron-phonon coupling matrix elements, atomic forces and total energies, charge densities, density matrices, and quasiparticle self-energies within the GW approximation.

Software ecosystem and computational workflow

As illustrated in Fig. 3, the typical research workflow within the DeepH framework proceeds sequentially from dataset generation and graph construction to model training, inference, and property post-processing. The platform supports a wide range of material systems, from bulk systems to complex defect structures, and offers multiple network architectures to balance accuracy and computational cost. To accommodate users across expertise levels, DeepH-pack implements a tiered interface framework progressing from accessibility to extensibility. First, a configuration-driven mode enables general users to deploy predefined architectures and training workflows through human-readable configuration files, positioning DeepH-pack as a versatile ecosystem for all-in-one computational materials solutions. Second, an object-oriented Python interface decomposes workflows into modular abstractions (data graphs, model blocks, training loops, etc.), facilitating the flexible assembly of custom architectures via class inheritance and composition. Finally, a low-level plugin system empowers developers to inject in-memory data across languages (Julia/C/Fortran) via MPI-based communication, enabling intervention in GPU training to modify intermediate network states (e.g., perturbing linear layer outputs) and seamless integration of external codes (e.g., DFT calculators), as demonstrated in the work of DeepH-Zero [41].

DeepH-pack utilizes the functional programming paradigm of JAX to implement efficient and differentiable electronic structure methods [56]. By leveraging the Accelerated Linear Algebra (XLA) compiler, the software achieves hardware-agnostic execution across diverse computing architectures, including CPUs, GPUs, and TPUs. The framework incorporates optimized Just-In-Time (JIT) compilation strategies to handle variable array shapes during runtime, which results in significant reductions in memory usage during gradient backpropagation and substantial increases in computational throughput. Complementing the core training package is DeepH-dock,

a companion toolkit designed to manage the end-to-end workflow [57]. While DeepH-pack focuses on the computation-intensive tasks of model optimization and inference, DeepH-dock handles the essential tasks of data preparation, bidirectional data conversion between internal formats and major DFT codes including SIESTA, Quantum ESPRESSO, FHI-aims, OpenMX, and ABACUS, and physical property post-processing.

Dataset construction and model training

In specialized applications, users often need to prepare custom datasets tailored to specific physical systems. For example, when modeling large supercells of amorphous materials, one may construct a training set from smaller representative amorphous structures. For moiré systems, a dataset can be built by generating untwisted layered van der Waals structures and then applying interlayer shifts and perturbations. Such training-set design strategies are discussed in further detail in the subsequent benchmarking and application sections. For training universal materials models, atomic structures are commonly sourced from established materials databases such as the Materials Project [58], ICSD [59], AFLOW [60], and Alexandria [61], which provide curated experimentally observed and computationally optimized configurations. Alternatively, traditional structure-search algorithms, including CALYPSO [62] and AIRSS [63] coupled with DFT software such as VASP [64], can be employed to systematically explore high-quality atomic configurations. More recently, modern generative neural networks like MatterGen [5], integrated with universal machine learning force fields, such as MatterSim [65] and CHGNet [66], have enabled efficient and broad exploration of structural spaces. Together, these approaches offer practical and accessible pathways to obtaining diverse, high-quality, and stable atomic structures for training machine learning models.

The electronic structure data are standardized into a unified format based on Hierarchical Data Format version 5 (HDF5) to ensure efficient storage and high-throughput parallel input/output performance (see Supplementary Information for detailed data structures). DeepH-dock facilitates parallelized conversion of output files from diverse DFT codes into this standard format. When given compatible atomic-orbital basis sets, DeepH-dock is able to project plane-wave (PW) DFT Hamiltonians onto atomic orbitals using the HPRO method proposed in [67], thereby enabling training on PW data. Furthermore, DeepH-dock incorporates a standalone module dedicated to the computation of overlap matrices implementing an efficient grid-based interpolation scheme [67] originally proposed by Sankey and Niklewski [68]. By exploiting the finite support of numerical atomic orbitals as well as efficient spline interpolations, the algorithm achieves linear scaling with respect to the system size. In our benchmark testing, this workflow computes the overlap matrix for an 11,164-atom magic-angle twisted bilayer graphene system (using SIESTA's DZP basis set) within two minutes on a single CPU core.

DeepH-pack handles diverse training scenarios, ranging from specialized datasets with a few hundred structures to large-scale materials databases containing hundreds of thousands of entries. The training pipeline supports flexible data management, allowing users to either load datasets entirely into CPU memory for speed or stream them from disk to handle massive datasets. Furthermore, the implementation supports efficient multi-GPU parallelism across single or multiple computing nodes. The training process begins by constructing a local coordinate graph representation of the crystal structure based on the sparsity pattern defined by the overlap matrix. In this graph, nodes represent individual atoms, while edges correspond to the non-zero Hamiltonian sub-blocks between atom pairs. Each node is assigned the atomic number of the corresponding atom, and each directed edge (i, j) carries the relative position vector $\mathcal{R}_{ij}$ from atom i to atom j . These graph features serve as the input to a JAX-based equivariant GNN, which preserves the physical symmetries of the underlying quantum interactions. For every connected pair of atoms i and j , the network outputs the predicted Hamiltonian matrix sub-block H_{ij} . The network parameters are optimized by minimizing a loss function $\mathcal{L}$ that quantifies the difference between the predicted Hamiltonian and the ground-truth labels. DeepH-pack allows users to select between Mean Squared Error (MSE) or Mean Absolute Error (MAE) as its optimization objective. Taking the MSE as an example, the loss for a batch of data is evaluated by:

$$\mathcal{L} = \frac{1}{N_b} \sum_{k=1}^{N_b} \frac{1}{|\mathcal{N}_k|} \sum_{(i,j) \in \mathcal{N}_k} \frac{1}{N_{ij}} \left\| \hat{H}_{ij}^{(k)} - H_{ij}^{(k)} \right\|^2, \quad (9)$$

where N_b denotes the batch size, $|\mathcal{N}_k|$ denotes the number of i - j pairs, N_{ij} denotes the total number of matrix elements in the matrix sub-block H_{ij} , $\mathcal{N}_k$ represents the set of interacting atom pairs in the k -th structure, and $\|\cdot\|$ denotes the Frobenius norm. After the training phase is complete, the optimized model is deployed to predict DFT Hamiltonians for unseen structures supporting high-throughput inference tasks efficiently utilizing multi-GPU acceleration.

Physical property post-processing

Once the Hamiltonian is predicted, DeepH-dock enables direct evaluation of fundamental electronic properties and offers bidirectional compatibility with established DFT workflows (detailed interface descriptions are provided in the Supplementary Information). The electronic band structure $\mathcal{E}_n(\mathbf{k})$ defined in Eq. (3) describes the single-particle energy levels of electrons in periodic crystals. For systems of moderate sizes (< 1000 atoms), DeepH-dock can directly solve the generalized eigenvalue problem in Eq. (3) to produce the band structure. In addition, DeepH-dock determines the Fermi energy according to the electronic occupation number and computes the electronic DOS from the energy levels. Figure 4 shows the band structure and DOS of bilayer black phosphorus obtained

in a post-processing step from a Hamiltonian predicted by DeepH-pack.

Wave functions describe the spatial distribution of electrons for each energy level and are required for many subsequent physical-property calculations, such as electronic conductivity [48], optical spectrum [49], and many-body perturbation theory corrections [69]. The Bloch wave function is defined by

$$\psi_{n\mathbf{k}}(\mathbf{r}) = \frac{1}{\sqrt{N}} \sum_{\mathbf{R}} \sum_{j\beta \in \text{uc}} e^{i\mathbf{k} \cdot \mathbf{R}} c_{n,j\beta}(\mathbf{k}) \phi_{j\beta}(\mathbf{r} - \mathbf{R} - \mathcal{R}_j), \quad (10)$$

where $\mathbf{R}$ denotes the lattice translation vector, and N is the size of the Born-von Karman supercell with atomic positions $\mathcal{R}_j$ restricted within the unit cell at the origin. After solving Eq. (3), DeepH-dock exports the wave function coefficients $c_{n,j\beta}(\mathbf{k})$ and computes the periodic part of the Bloch wave function in real space via

$$u_{n\mathbf{k}}(\mathbf{r}) = \frac{1}{\sqrt{N}} \sum_{\mathbf{R}} \sum_{j\beta \in \text{uc}} e^{-i\mathbf{k} \cdot (\mathbf{r} - \mathbf{R})} c_{n,j\beta}(\mathbf{k}) \phi_{j\beta}(\mathbf{r} - \mathbf{R} - \mathcal{R}_j). \quad (11)$$

Furthermore, DeepH-dock can compute the one-particle density matrix under the atomic orbital basis, defined as

$$\rho_{i\alpha,j\beta}(\mathbf{k}) = \sum_n f_{n\mathbf{k}} c_{n,i\alpha}(\mathbf{k}) c_{n,j\beta}^*(\mathbf{k}), \quad (12)$$

where i, j index atoms in the unit cell, α, β label orbitals on each atom, $f_{n\mathbf{k}}$ denotes the Fermi-Dirac occupation factor of state $(n, \mathbf{k})$, and the summation runs over all bands.

Benchmarking

To rigorously evaluate the accuracy and efficiency of DeepH-pack, we employ a comprehensive set of benchmarks spanning diverse dimensionalities and chemical environments. The test systems range from molecular systems (water molecule) to two-dimensional (2D) materials (monolayer MoS₂, bilayer Bi₂Se₃) and bulk crystals (graphite, SiC, and carbon allotropes). Crucially, these datasets are selected to address critical physical challenges, including relativistic SOC effects in Bi₂Se₃, defect states in SiC, and structural polymorphism in carbon. The performance of the networks implemented in DeepH-pack, namely the **Normal** and **Accurate** architectures, is evaluated, including training/inference time and memory usage. A comparison with the previous DeepH-E3 implementation is also provided. All training and inference processes were performed on NVIDIA RTX 4090 GPUs with AMD EPYC 7502 Processors. The model accuracy is assessed based on a set of standardized metrics listed below. The error between the predicted Hamiltonian $H_{i\alpha,j\beta}$ and the benchmark result is evaluated as the most direct measurement of model accuracy. The accuracy of observable quantities is also evaluated, including the mean absolute error (MAE) in band energies ($\mathcal{E}_{n(\mathbf{k})}$), the MAE in DOS ($D(\epsilon)$), and the root mean square error (RMSE)

of eigenstates ($\psi_{n(\mathbf{k})}$) at Γ point. The metrics are defined below.

$$\text{MAE}_H = \frac{\sum_{i\alpha,j\beta} |H_{i\alpha,j\beta}^{\text{pred}} - H_{i\alpha,j\beta}^{\text{bm}} - \Delta\mu_H S_{i\alpha,j\beta}|}{N_H}, \quad (13)$$

$$\text{MAE}_{\text{band}} = \frac{\int d^n \mathbf{k} \sum'_n |\mathcal{E}_{n(\mathbf{k})}^{\text{pred}} - \mathcal{E}_{n(\mathbf{k})}^{\text{bm}} - \Delta\mu_{\text{band}}|}{\int d^n \mathbf{k} N'_{\text{band}(\mathbf{k})}}, \quad (14)$$

$$\text{MAE}_{\text{DOS}} = \frac{\int_{\varepsilon_{\min}}^{\varepsilon_{\max}} |D_{(\varepsilon+\Delta\mu_{\text{band}})}^{\text{pred}} - D_{(\varepsilon)}^{\text{bm}}| d\varepsilon}{\int_{\varepsilon_{\min}}^{\varepsilon_{\max}} D_{(\varepsilon)}^{\text{bm}} d\varepsilon}, \quad (15)$$

$$\text{RMSE}_{\psi,\mathbf{k}=\mathbf{0}} = \left(\frac{\sum'_n (1 - \sum''_g |\langle \psi_{g(\mathbf{k}=\mathbf{0})}^{\text{bm}} | \psi_{n(\mathbf{k}=\mathbf{0})}^{\text{pred}} \rangle|^2)}{N'_{\text{band}(\mathbf{k}=\mathbf{0})}} \right)^{1/2}. \quad (16)$$

These observable-based metrics are essential because they directly assess the model's performance on physical properties, rather than relying solely on the accuracy of the Hamiltonian matrix (see also Supplementary Information, Section S7).

The reciprocal space is sampled sufficiently to evaluate the energy bands and DOS. For water molecules, only the Γ point is included. For periodic crystals, the Brillouin zones are sampled with a relatively dense grid with a $\mathbf{k}$ -point spacing of $0.02\text{--}0.07\text{\AA}^{-1}$. Gaussian smearing is applied to evaluate DOS with a smearing width of $0.01\text{--}0.08$ eV. To extract information near the Fermi level, the errors of the energy bands, DOS, and eigenstates are calculated within an energy window $(\varepsilon_{\min}, \varepsilon_{\max})$. This is indicated by $\sum'$, which means $\sum'_n = \sum_n, \text{ s.t. } \varepsilon_{\min} < \mathcal{E}_{n(\mathbf{k})}^{\text{bm}} < \varepsilon_{\max}$. The energy window is initially centered at the Fermi energy ε_F with a total width of 6.0 eV, and is expanded until at least 10 energy bands are included (except for water molecules, where the 4 occupied and the 7 lowest unoccupied energy levels are included). In the denominator, N_H is the total number of elements in the Hamiltonian matrix, and $N'_{\text{band}(\mathbf{k})}$ is the number of energy levels within the energy window. The note $\sum''$ in the expression of RMSE_{ψ} denotes the summation restricted within the nearly degenerate subspace related to ψ_n , which is identified by an energy spacing threshold ε_{th} (defaulting to 5 meV). The inner product between eigenstates is calculated as $\langle \psi_{m(\mathbf{k})} | \psi_{n(\mathbf{k})} \rangle = c_{m(\mathbf{k})}^\dagger S(\mathbf{k}) c_{n(\mathbf{k})}$, where $S(\mathbf{k})$ is the Fourier transform of the overlap matrix and $c_{n(\mathbf{k})}$ is the eigenvector from diagonalization. The errors of the Hamiltonian matrix, band energies, and DOS are defined after gauge correction $\Delta\mu$ [40].

$$\Delta\mu_H = \frac{\sum_{i\alpha,j\beta} (H_{i\alpha,j\beta}^{\text{pred}} - H_{i\alpha,j\beta}^{\text{bm}}) S_{i\alpha,j\beta}}{\sum_{i\alpha,j\beta} S_{i\alpha,j\beta}^2} \quad (17)$$

$$\Delta\mu_{\text{band}} = \text{median}\{\mathcal{E}_{n(\mathbf{k})}^{\text{pred}} - \mathcal{E}_{n(\mathbf{k})}^{\text{bm}} \mid \varepsilon_{\min} < \mathcal{E}_{n(\mathbf{k})}^{\text{bm}} < \varepsilon_{\max}\} \quad (18)$$

The results are shown in Tab. 1, and the best-performing values for each metric are highlighted in bold.

We select water as a representative of small molecules. The dataset contains 4999 configurations of water molecules from the MD17 dataset [70], which consists of structures sampled from molecular dynamics trajectories. The **Normal** and **Accurate** models converged after 8 hours of training, with their performance evaluated on the same test set. Both **Normal** and **Accurate** models outperform DeepH-E3, achieving inference speeds 3 to 5 times faster and reducing errors by 50-70%. The **Accurate** model predicts the most accurate Hamiltonian matrix and energy levels, while the **Normal** model achieves a comparable accuracy with much fewer parameters and less inference time. Only the 4 occupied and 7 lowest unoccupied states are included in the band energy MAE and eigenstate RMSE analysis, as higher unoccupied states are not practically relevant to most of the property calculations and experiments. It should be noted that the software and calculation setup in this work are different from the published MD17 dataset, and the Hamiltonian matrix error cannot be directly compared with other works [13, 26].

Monolayer MoS_2 and bilayer Bi_2Se_3 were selected as representative 2D materials. Both are van der Waals materials and can be grown into thin layers [71, 72]. These datasets have been used in the work of DeepH-E3 [24]. MoS_2 belongs to the transition metal dichalcogenide (TMD) family and shows outstanding electronic and optical qualities, notably high carrier mobility and strong excitonic effects [73–75]. These properties enable its potential applications in transistors and photodetectors. The dataset is constructed by 5×5 supercells with random perturbation of atom positions, resulting in a total of 500 structures. DeepH-pack demonstrates robust performance on these datasets. The **Normal** network achieves significantly higher accuracy (reducing MAE by $\sim 70\%$) than DeepH-E3 with the smallest time cost in training and inference processes. The **Accurate** network further improves the accuracy by approximately 3-fold with lower memory consumption than **Normal** network. The four accuracy metrics show the same trend that **Accurate** network performs the best, **Normal** network in the middle, both better than DeepH-E3.

Bi_2Se_3 is known by its high thermoelectric efficiency and non-trivial topological properties [76], and is a typical material with significant SOC effects. The dataset consists of 576 structures, each constructed as a 3×3 supercell with in-plane interlayer shifts between the two layers and random atomic perturbations of $0.1\text{\AA}$ applied along each spatial direction. SOC was included in the DFT calculations because it plays an essential role in the band inversion and leads to the non-trivial band topology. To handle the additional spin angular momentum of $1/2 \otimes 1/2 = 0 \oplus 1$, the maximum angular momentum (L) in the network needs to be 1 plus twice the maximum angular quantum number of the basis set orbitals, that is, $L = 2 \times 2 + 1 = 5$ in this case. **Normal** and **Accurate** networks demonstrate comparable overall performance on this dataset. Specifically, **Accurate** network achieves smaller Hamiltonian MAE, while **Normal** network

Table 1 The efficiency and accuracy of DeepH-pack on benchmark datasets. The bold items mark the best result across tested models.

		Water molecule	Monolayer MoS ₂	Bilayer Bi ₂ Se ₃	Graphite	Bulk Cu	Defective SiC	Carbon allotropes
Number of atoms per unit cell		3	75	90	108	32	64	3-58
Training set size		3000	300	460	240	180	300	342
Batch size		30	1	1	1	1	1	1
Hamiltonian matrix MAE (meV) ↓	DeepH-E3 [24]	0.25	0.43	0.27	0.36	0.36	0.20	1.8
	Normal	0.11	0.13	0.077	0.21	0.086	0.058	0.69
	Accurate	0.077	0.041	0.044	0.13	0.039	0.037	0.53
Band energy MAE (meV) ↓	DeepH-E3	2.4	0.74	0.97	12	0.86	0.50	45
	Normal	1.3	0.22	0.29	12	0.15	0.24	17
	Accurate	0.71	0.060	0.31	8.7	0.095	0.20	10
Density of states MAE ($\times 10^{-3}$, dimensionless) ↓	DeepH-E3	–	8.5	1.8	17	3.2	0.61	27
	Normal	–	2.4	0.65	16	0.28	0.27	17
	Accurate	–	0.70	0.70	12	0.18	0.21	11
Eigenstate RMSE ($\times 10^{-2}$, dimensionless) ↓	DeepH-E3	0.22	7.2	5.1	7.0	7.3	2.0	11
	Normal	0.13	1.8	2.9	7.9	2.2	0.76	5.2
	Accurate	0.12	0.61	3.5	3.9	1.5	0.70	2.2
Training time (s/batch) ↓	DeepH-E3	0.63	1.0	1.3	0.63	0.66	1.1	0.92
	Normal	0.052	0.12	0.23	0.14	0.08	0.21	0.23
	Accurate	0.076	0.19	0.23	0.23	0.12	0.31	0.17
Inference time (s/structure) ↓	DeepH-E3	0.12	0.17	0.24	0.11	0.12	0.21	0.19
	Normal	0.02	0.03	0.05	0.04	0.02	0.06	0.06
	Accurate	0.03	0.07	0.13	0.13	0.05	0.14	0.08
Inference memory (MB/atom) ↓	Normal	0.9	7.9	13	6.5	17	13	22
	Accurate	0.9	6.4	9.1	5.9	10	12	7.9
Parameter size (million)	DeepH-E3	0.68	0.77	1.46	0.42	0.64	0.72	0.84
	Normal	1.36	2.06	1.29	0.57	1.49	1.98	3.48
	Accurate	3.58	8.39	7.52	3.64	6.34	8.39	2.62

yields slightly smaller error in energy bands, DOS, and eigenstates. Even for the heavy **Accurate** model with 7-8 million parameters, the training on the two datasets can be completed within 48 hours, which greatly facilitates the hyperparameter optimization and accelerates the research works.

Graphite, bulk Cu, SiC, and carbon allotropes were chosen to represent bulk systems. Graphite has a layered structure analogous to the MoS₂ mentioned above, and is representative of 3D van der Waals materials. Furthermore, a variety of stacking modes (e.g. AA, AB, ABC) dictate diverse mechanical properties and electronic structures, for instance, the strong correlation effects in rhombohedral graphite [77, 78]. The dataset is constructed by $3 \times 3 \times 6$ cell expansion of the AB stacked graphite. Random perturbations of atom positions are then added to give 300 distinct structures. Bulk Cu is included as a representative elemental metal to demonstrate the model’s robustness on metallic systems. The Cu dataset is prepared through the initial sampling of diverse atomic configurations via molecular dynamics simulations, followed by first-principles calculations using OpenMX,

yielding 300 distinct structures. Comprehensive benchmark results on additional elemental metals (e.g., Al and Au) are detailed in the Supplementary Information. Silicon carbide (SiC), a semiconductor possessing wide band gap and large thermal conductance, is chosen as a representative of materials with defects [37]. To address the challenge posed by the intrinsic point defects on device performance, a dataset of 500 structures is generated to model the anti-site defects. Each structure is created by random replacement between carbon and silicon atoms in a $2 \times 2 \times 2$ supercell. The relative performance between the networks on the graphite and SiC dataset remains consistent with prior observations on the MoS₂ dataset, whereas **Normal** network outperforms in time cost and **Accurate** network outperforms in model accuracy. Note that the memory cost per atom for the SiC system is larger than the MoS₂ dataset, because the number of neighbors in bulk materials (~ 130 for the SiC dataset) is larger than that in 2D materials (~ 60 for monolayer MoS₂). The last dataset contains 427 carbon allotropes from the Samara Carbon Allotrope Database [79], all of which are stable

Table 2 The ablation study of **Accurate** network on benchmark datasets. The **Flexible** setting has fewer channels within higher-order angular momentum and the **Uniform** setting has the same number of channels within all the angular momenta. The bold items mark the best result across tested models.

		Water molecule	Monolayer MoS ₂	Bilayer Bi ₂ Se ₃	Graphite	Bulk Cu	Defective SiC	Carbon allotropes
Hamiltonian matrix MAE (meV) ↓	Flexible	0.077	0.041	0.044	0.13	0.039	0.037	0.53
	Uniform	0.064	0.023	0.044	0.13	0.041	0.033	0.54
Band energy MAE (meV) ↓	Flexible	0.71	0.060	0.31	8.7	0.095	0.20	10
	Uniform	0.60	0.039	0.37	9.3	0.088	0.17	11
Density of states MAE ($\times 10^{-3}$, dimensionless) ↓	Flexible	–	0.70	0.70	12	0.18	0.21	11
	Uniform	–	0.43	0.87	12	0.16	0.19	12
Eigenstate RMSE ($\times 10^{-2}$, dimensionless) ↓	Flexible	0.12	0.61	3.5	3.9	1.5	0.70	2.2
	Uniform	0.075	0.37	3.9	5.1	1.4	0.67	2.9
Training time (s/batch) ↓	Flexible	0.076	0.19	0.23	0.23	0.12	0.31	0.17
	Uniform	0.076	0.28	0.43	0.46	0.16	0.51	0.21
Inference time (s/structure) ↓	Flexible	0.03	0.07	0.13	0.13	0.05	0.14	0.08
	Uniform	0.03	0.10	0.18	0.21	0.06	0.20	0.09
Inference memory (MB/atom) ↓	Flexible	0.9	6.4	9.1	5.9	10	12	7.9
	Uniform	2.3	13	19	15	17	23	14
Parameter size (million)	Flexible	3.6	8.4	7.5	3.6	6.3	8.4	2.6
	Uniform	10.4	24.5	29.8	18.4	18.4	24.5	7.7

structures. The dataset sparsely samples a large configurational space with a limited number of data points, which poses a significant challenge for model generalization. In this complicated case study, **Accurate** network exhibits expressivity and generalization capabilities superior to those of the other networks. The **Accurate** network achieves a remarkably low band MAE of 10 meV and relative errors of approximately 0.01 for the DOS and 0.02 for the eigenstates. These high-accuracy results strongly support the reliability of subsequent property calculations (e.g., optical and response properties) derived from the predicted Hamiltonian. Training on the graphite dataset requires 60 hours to reach full convergence. However, by applying a smaller patience in the learning rate scheduler, the time cost can be considerably reduced with an accuracy drop of less than 10%. For the SiC and carbon allotropes datasets, the training processes are completed within 48 hours, which is completely acceptable.

To explicitly validate the efficacy of the channel-flexible eSCN algorithm, we conducted an ablation study on the **Accurate** network. The performance of two parameter settings, namely the **Flexible** and **Uniform**, are compared with each other in Tab. 2. The two settings share the same architecture of the **Accurate** model, and the only difference is that the **Flexible** setting has fewer channels within higher-order angular momentum (e.g., $64\times 0e+32\times 1e+16\times 2e+8\times 3e+4\times 4e$), while the **Uniform** setting enforces a uniform channel width across all angular momentum orders l , matching the channel count of $l = 0$ (e.g., $64\times 0e+64\times 1e+64\times 2e+64\times 3e+64\times 4e$). As demonstrated in Tab. 2, benefited from the channel-flexible design, the **Flexible** setting drastically reduces the total parameter count by approximately 60% to 75% compared to the **Uniform**. Consequently, this structural

streamlining translates to significant computational savings: the **Flexible** setting exhibits substantially faster training and inference speeds while cutting the inference memory footprint per atom by up to 50% (Fig. S2). Crucially, these massive gains in efficiency do not compromise predictive performance. Except for a deviation observed in the monolayer MoS₂ dataset, the **Flexible** setting achieves almost identical accuracy to the computationally prohibitive **Uniform** across all evaluated metrics. In the carbon allotropes dataset, the **Flexible** setting even performs slightly better than the **Uniform**, because the latter contains an excessive number of parameters and suffers from more severe overfitting problems. These ablation tests rigorously confirm that the channel-flexible eSCN module successfully eliminates redundant parameters associated with high-order representations without sacrificing the underlying representational capacity of the network. Consequently, the **Flexible** setting provides an optimized balance between efficiency and accuracy, making it our strongly recommended configuration for most scenarios (see Supplementary Information, Section S5). Furthermore, successfully overcoming the rigid constraints of the traditional **Uniform** setting exemplifies the fundamental algorithmic innovations empowered by our architectural reconstruction.

Benefiting from the optimized code engineering in **e3nn-jax** [80] and DeepH-pack, the **Normal** network, which has the same architecture as DeepH-E3, outperforms DeepH-E3, reducing errors by $\sim 40\%$ to $\sim 70\%$ and increasing efficiency by 100% to 800%. The **Accurate** network achieves even $\sim 20\%$ to $\sim 40\%$ lower error due to advanced architecture design and larger parameter size, especially on general-purpose datasets such as the carbon allotropes dataset. Designed for large-capacity general-purpose material modeling, the **Accurate** model achieves

greater memory efficiency than the **Normal** model through its use of a semi-local coordinate frame, thus enabling larger-scale training.

Scalability and application scope

Traditional density functional theory faces severe computational bottlenecks when applied to systems with extremely large dimensions or complex chemical compositions. DeepH-pack overcomes these fundamental limitations through the scalability of its software architecture and the predictive power of its equivariant neural networks. By maintaining DFT accuracy while achieving linear-scaling efficiency, our framework makes previously intractable calculations highly feasible. To demonstrate this capability, we present three representative applications. These examples highlight how DeepH-pack enables the study of systems at length scales directly relevant to real-world experimental settings, such as massive moiré superlattices in twisted two-dimensional materials and extended structural models of amorphous silicon. Furthermore, we demonstrate the capacity of the framework to handle highly diverse chemical spaces through the development of a universal materials foundation model. Ultimately, this computational paradigm empowers researchers to explore a significantly richer spectrum of emergent condensed matter physics phenomena that remain inaccessible to conventional first-principles methods.

Moiré superlattices in twisted TMDs

Twisted bilayer transition metal dichalcogenides (TMDs) have emerged as a prominent research focus in condensed matter physics and materials science [81–84]. By precisely controlling the interlayer twist angle, their electronic structures can be effectively engineered, giving rise to a rich variety of correlated electronic states and topological properties. Among them, twisted bilayer MoTe₂ has attracted significant attention due to the potential emergence of correlated insulating states, topologically non-trivial bands, and possible superconducting behavior near specific “magic” angles, making it an ideal platform for exploring 2D strongly correlated physics and topological quantum phenomena. However, performing full first-principles calculations on such systems is highly challenging because the size of the moiré superlattice increases dramatically as the twist angle decreases. The computational cost of conventional DFT scales cubically with the number of atoms, rendering systematic angle-dependent scanning computationally prohibitive. Therefore, developing approaches that retain first-principles accuracy while achieving high computational efficiency is particularly urgent. This creates an important application scenario for machine learning methods, especially the DeepH framework, which can efficiently predict the DFT Hamiltonians directly from atomic structures. By constructing a highly transferable Hamiltonian model, DeepH holds the potential to enable high-throughput computation and rapid screening across multiple degrees of freedom such as twist

angle and stacking order, thereby systematically revealing the structure-property relationships in moiré materials.

High-accuracy prediction of Hamiltonians for supercell structures with DeepH typically begins by training a specialized model on a set of smaller-scale configurations. Since DeepH directly predicts Hamiltonian matrix elements between atom pairs, knowledge of local interactions learned from smaller systems can be effectively transferred to much larger structures, enabling rapid electronic-structure prediction at DFT accuracy. The selection of training structures is crucial when building a specialized model. The guiding principle is that the bonding environments present in the training set should comprehensively cover those appearing in the target large system. To demonstrate DeepH’s capability to generalize from small to large scales, we select a relaxed twisted bilayer MoTe₂ system containing 1,986 atoms as the prediction target. SOC is neglected here to keep the reference DFT calculation computationally tractable. To illustrate the significant impact of training-set design on model performance, we compare two different training-set generation strategies (Fig. 5(a)). Strategy I establishes a baseline dataset comprising 304 bilayer MoTe₂ structures divided into two categories. The first category consists of 256 supercells with 54 atoms, generated by systematically sampling in-plane interlayer translations on a uniform 16×16 grid. For these structures, the interlayer distance is uniformly sampled between 6.9 Å and 7.9 Å to capture the interlayer corrugation inherent in the relaxed twisted bilayer, and random atomic displacements within a range of ± 0.1 Å are applied to enhance structural diversity. The second category targets the specific stacking motifs found in twisted systems. It includes 48 structures derived from relaxed twisted bilayers with unit cell sizes of 42, 114, and 222 atoms, which naturally cover mixed AA and AB stacking regions. Inspired by the MLIP validation protocol introduced in Ref. [85], strategy II extends Strategy I by incorporating an additional set of 64 quasi-1D moiré structures, each containing 93 atoms. These structures are constructed by applying uniaxial strain to one layer relative to the other. This construction method allows the formation of rich moiré interference patterns within a supercell of moderate size, enabling the model to learn complex long-range modulations efficiently. Both strategies employed the same model architecture and hyperparameters: an embedding layer with 8.0 Å Gaussian smearing, an intermediate representation of $64 \times 0e + 48 \times 1e + 32 \times 2e + 32 \times 3e + 16 \times 4e$, and a 4-layer graph neural network with 2 attention heads per layer, resulting in 12,099,170 trainable parameters. All models were trained in FP32 precision on a single RTX 4090 GPU using a batch size of 1, the AdamW optimizer (initial learning rate = 0.001, $\beta = (0.9, 0.999)$), and a ReduceLROnPlateau scheduler with a patience of 60 epochs. For Strategy I, the 304 available structures were split into 228 for training, 38 for validation, and 38 for testing. Training converged after 1,111 epochs (11.1 hours), achieving final MAE of 0.091 meV on the training set, 0.105 meV on the validation set, and 0.085 meV on the test set. In contrast, Strategy II used 368 structures partitioned into

276 for training, 46 for validation, and 46 for testing. This model converged after 1,148 epochs (15.6 hours), with final MAEs of 0.084 meV (training), 0.115 meV (validation), and 0.120 meV (test).

For the 1,986-atom twisted bilayer MoTe₂, Strategy II significantly improves model accuracy over Strategy I: the MAE of the Hamiltonian matrix elements decreases from 0.42 meV to 0.29 meV. This improvement is further reflected in band-structure predictions. As shown in Fig. 5 (b,c), the band MAEs are 4.64 meV and 1.26 meV for Strategy I and Strategy II, respectively, with the accuracy of moiré flat bands showing particularly notable gains. The elimination of the deviations observed in Fig. 5(b) by the enhanced dataset in Fig. 5(c) attests to the high expressivity of the DeepH model. This contrast suggests that performance limitations are often attributable to data coverage rather than model capacity; therefore, systematically enriching the training set with representative local environments offers a reliable way to resolve potential accuracy bottlenecks.

Electronic structure of large-scale amorphous silicon

Amorphous silicon (a-Si) serves as a prototypical model for the extensive family of non-crystalline semiconductors, finding critical applications in thin-film transistors and solar cells [86, 87]. Unlike its crystalline counterpart, a-Si lacks long-range translational symmetry but retains a rigid short-range tetrahedral network. This structural duality presents a unique challenge for computational modeling. While DFT provides accurate descriptions of local bonding, the study of disorder-induced phenomena requires large supercells containing thousands of atoms to obtain statistically meaningful results. Such scales are often computationally intractable for standard DFT, hindering systematic investigations of structure-property relationships. However, since the electronic structure of a-Si is predominantly governed by local chemical environments and short-range covalent interactions, this material represents an ideal candidate system for applying the DeepH method. By accurately capturing this local physics from moderate-sized training samples, DeepH offers a scalable solution that maintains quantum-mechanical accuracy, effectively opening the door to efficient electronic-structure calculations for the broader class of amorphous and disordered materials.

To demonstrate this capability, we established a workflow to model the electronic properties of large-scale a-Si systems. To train a DeepH model, we first sample small a-Si structures via a melt-quench molecular dynamics procedure using the LAMMPS package [88] for training set construction. 300 independent a-Si configurations were generated from this process, each containing 216 atoms, to ensure a comprehensive sampling of the diverse local atomic environments found in the disordered network. For the training data, we performed self-consistent DFT calculations on these structures using Quantum ESPRESSO, and the resulting Hamiltonians were then projected from the PW basis onto SIESTA's atomic-orbital basis set. As illustrated in Fig. 6(a), these labeled datasets were

used to train the DeepH model. The network, based on the **Accurate** architecture, was trained on the dataset. The 300 structures were split into 240 for training, 30 for validation, and 30 for testing. The model was trained using FP32 precision over 11.3 hours on four RTX 4090 GPUs. Key hyperparameters include: an embedding Gaussian smearing of 7.0 Å, an intermediate representation of 48x0e+48x1e+40x2e+16x3e+8x4e, a 4-layer GNN with 2 attention heads per layer, resulting in a total of 6,073,554 parameters. We used a batch size of 4 (four materials per batch), the AdamW optimizer with an initial learning rate of 0.001 and β parameters = (0.9, 0.999), and a ReduceLROnPlateau scheduler with a patience of 60 epochs and reduction factor of 0.5. The model was trained for 377 epochs, achieving a final training MAE of 0.58 meV, a validation MAE of 0.99 meV, and a test loss of 0.94 meV in Hamiltonian matrix elements.

Once trained, the model was deployed to predict the Hamiltonian of a significantly larger supercell containing 4096 atoms. This large-scale structure was adopted from Ref. [89]. To validate the physical accuracy of the prediction, we analyzed the spectral properties of this large-scale system. Figure 6(b) displays the predicted DOS, and Fig. 6(c) presents the effective band structure obtained via the band unfolding technique using SIESTA [45, 90]. The results successfully reproduce the characteristic features of the a-Si electronic structure, including disorder-induced broadening of energy bands, thereby confirming the model's robust transferability and its ability to bridge the scale gap in amorphous materials simulation. Our workflow integrates a series of computational tools: LAMMPS for structure generation, Quantum ESPRESSO for training-set production, DeepH-pack for model training, and SIESTA for post-processing. It demonstrates the flexibility of the DeepH framework in enabling seamless interoperability across software ecosystems. This effort highlights a step toward building a more unified and cohesive computational materials science community.

Universal materials model

Accurate determination of electronic structures is fundamental to understanding the physicochemical properties of materials at the microscopic level. While traditional first-principles methods offer high accuracy, their steep scaling with system size often imposes prohibitive computational costs, thereby limiting their applicability in high-throughput screening and large-scale simulations. The previously presented applications exemplify the “specialized model” approach within the DeepH framework. These models are tailored to specific material systems, enabling high-precision investigations of intricate physical properties driven by complex structural variations within a fixed compositional space. In parallel with these focused applications, a universal materials model of electronic structure represents a complementary paradigm that prioritizes broad transferability. It aims to establish a generalizable mapping from atomic configurations to electronic Hamiltonians across arbitrary elemental combinations. Consequently, a single trained model can not

only provide rapid predictions for diverse materials but also serve as a robust pre-trained foundation that can be efficiently fine-tuned for downstream tasks. Assuming sufficient accuracy and chemical coverage, such a model has the potential to fundamentally transform the landscape of computational materials science. By circumventing the need for repetitive *ab initio* calculations, it facilitates high-throughput virtual screening across vast chemical spaces and serves as an efficient engine for inverse materials design and simulations of complex defect or interface systems. Ultimately, this approach provides a transformative computational tool to accelerate the discovery and design of novel materials.

The DeepH algorithm offers comprehensive support for training universal materials models of electronic structure. As a demonstration of this capability, we constructed an example universal materials model applicable to elements from the first four periods, focusing on non-magnetic systems without SOC. The training data were sourced from the Materials Project database, from which we selected 9,510 non-magnetic crystal structures composed of elements from the first four periods. The corresponding DFT Hamiltonian labels were calculated using the computational science workflow manager AiiDA [91, 92] in combination with OpenMX [47]. Details on the dataset configuration can be found in the DeepH universal materials model publication [93]. This dataset exhibits high diversity in element types, stoichiometry, crystal structure types, and bonding characteristics, ensuring the model can adequately learn the complex relationships between inter-atomic interactions and the electronic Hamiltonian across varied chemical environments [93].

The core architecture of the model is specifically designed to uniformly handle different element types and atomic systems of variable sizes. Its learning objective is to directly infer the DFT Hamiltonian from the input atomic coordinates and element types, from which various electronic properties can be derived. The network, based on the **Accurate** architecture, was trained on the dataset. The 9,510 structures were split into 5,692 for training, 1,927 for validation, and 1,891 for testing. The model was trained using FP32 precision over 99.6 hours on four RTX 4090 GPUs. Key hyperparameters include: an embedding Gaussian smearing of 9.0 Å, an intermediate representation of $96 \times 0e + 96 \times 1e + 80 \times 2e + 40 \times 3e + 20 \times 4e + 8 \times 5e$, a 6-layer GNN with 2 attention heads per layer, resulting in a total of 47,761,367 parameters. We used a batch size of 1 (one material per batch), the AdamW optimizer with an initial learning rate of 0.0004 and β parameters = (0.9, 0.999), and a WarmupAndCosineDecay scheduler with 10,000 warmup steps. The model was trained for 151 epochs, achieving a final training MAE of 0.59 meV, a validation MAE of 1.27 meV, and a test loss of 1.31 meV in Hamiltonian matrix elements. The observed overfitting suggests there is room for further model improvement. By training on this large-scale, diverse dataset, the model does not merely memorize specific material instances. Instead, it internalizes the underlying physical rules connecting atomic numbers, geometric configurations, and quantum mechanical operators,

thereby acquiring the capability to generalize and predict properties for previously unseen material structures.

To validate the effectiveness and demonstrate the strong potential of the constructed universal materials model, we applied it to perform predictions on a series of unseen material systems with diverse structures. This universal materials model successfully predicts the DFT Hamiltonians for materials spanning the first four periods of the periodic table across various morphologies. As shown in Fig. 7(a), the model achieves a prediction accuracy of better than 2 meV for all elements included in the test set. Figure 7(b) demonstrates that the mean prediction error across the test set is as low as 1.31 meV. The predicted results exhibit excellent agreement with first-principles benchmark calculations, which confirms that the model has successfully captured the universal governing laws of electronic structures across disparate materials. This work demonstrates that the deep-learning-based universal materials model of electronic structure has reached a level of practical utility. Its capability to generalize to a broad spectrum of material systems without structure-specific retraining lays a solid foundation for building the next-generation intelligent computational platform for materials science, marking a crucial step toward the ultimate goal of efficient, accurate, and comprehensive materials design and discovery.

Discussion

This work presents DeepH-pack, a unified software toolkit that bridges first-principles methods with deep learning to enable accurate, efficient, and generalizable electronic structure calculations. By integrating fundamental physical principles (such as locality and equivariance) into neural-network architectures, this toolkit provides a robust foundation for bridging the gap between advanced deep learning methods and practical, high-throughput materials discovery workflows. Compared to prior independent and fragmented implementations within the DeepH framework, DeepH-pack delivers tangible advantages across three primary dimensions. First, the complete backend re-engineering under the JAX paradigm and XLA compilation provides a leap in computational efficiency, enabling a 4-to-12 times acceleration in training throughput and hardware-agnostic deployment. Second, the network customization design introduces situational adaptability. The lightweight **Normal** network maximizes inference speed for routinely perturbed molecular and crystal tasks, while the channel-flexible **Accurate** network successfully overcomes the $O(L^6)$ complexity scaling bottleneck of traditional equivariant convolutions, and at the same time cutting redundant high-order tensor parameters by up to 75% without accuracy loss. Third, the seamless co-development with the companion toolkit, DeepH-dock, establishes exceptional ecosystem interoperability. By functioning as a standardized, bidirectional data interface compatible with multiple major DFT software suites, it lowers the application barrier for general users. Furthermore, its post-processing utilities empower researchers to systematically map deep-learning predicted

Hamiltonians back to physical quantities, transforming the framework into an end-to-end platform for practical atomistic simulations.

The development of DeepH-pack builds upon foundational deep-learning electronic structure methods [9], systematically evolving toward greater intelligence and generality. Core algorithmic advances include equivariant neural networks [24, 25], transformer-based universal models [40, 93], and physics-informed unsupervised learning [41]. Concurrently, its physical capabilities have been significantly expanded to encompass density functional perturbation theory [39], hybrid functionals [38], density-matrix learning [19], and plane-wave interfaces [67]. Driven by both our team and the broader community, the framework has been successfully applied to diverse complex phenomena, including moiré superlattices [32, 94], non-adiabatic dynamics [31], complex nanotubes [33], and defect systems [34], alongside continuously growing program interfaces [35].

Despite this expansive theoretical landscape, the current production-ready release of DeepH-pack is limited to standard DFT-level calculations and does not yet natively incorporate advanced modules such as density functional perturbation theory (DFPT). Nevertheless, the DeepH methodology is fundamentally general and can be trained to learn diverse physical operators beyond ground-state DFT. By leveraging symmetry-adapted tensor transformations, future extensions will target higher-level many-body perturbation methods and excited-state phenomena, encompassing the DFPT, *GW* approximation [69], the Bethe-Salpeter equation (BSE) [50], and time-dependent DFT (TDDFT) [95]. Another pivotal direction is the development of an electronic-structure foundation model. Through training on large datasets, such a model would provide universal predictions for arbitrary material structures, thereby laying the groundwork for large materials models. Concurrently, DeepH-pack will be continuously refined to better serve the community, with focused improvements in usability, interoperability, and computational scalability. This ongoing development aims to empower researchers by facilitating a seamless transition into the AI-driven era of computational physics. Ultimately, by making high-accuracy electronic structure modeling simultaneously efficient, physically complete, and accessible, DeepH-pack strives to fundamentally reshape the materials research ecosystem, accelerating the discovery and design of functional materials across scales.

Methods

Training data

To demonstrate the flexibility of DeepH-pack’s workflow support, DFT and MD calculations in training set generation processes were executed using multiple software platforms. Unless otherwise specified, a Monkhorst-Pack $\mathbf{k}$ -point sampling grid was employed for the DFT calculations of periodic systems. The number of $\mathbf{k}$ -points $N_{\mathbf{k}i}$ along each dimension was chosen such that the product

$|\mathbf{a}_i| \cdot N_{\mathbf{k}i}$ was closest to $80\text{\AA}$, where $\mathbf{a}_i$ denotes the lattice vector in that dimension. For 2D systems, the $\mathbf{k}$ -point sampling in the out-of-plane direction was restricted to 1.

Calculations for the water molecule dataset were performed using the AI2DFT code [41], employing norm-conserving (NC) pseudopotentials [96] and a double-zeta polarized (DZP) basis set [97]. The generalized gradient approximation (GGA) parameterized by Perdew, Burke, and Ernzerhof (PBE) was utilized for the exchange-correlation functional [98].

For monolayer MoS_2 , bilayer Bi_2Se_3 , graphite, bulk Cu, and carbon allotropes, the DFT calculations were conducted using OpenMX [47] with NC pseudopotentials [99], the **standard** multiple-zeta polarized basis set [100], and the PBE functional. SOC was explicitly incorporated for bilayer Bi_2Se_3 to accurately capture the topological band inversions.

For defective SiC, the electronic structures were computed using the SIESTA code [45] with NC pseudopotentials, a DZP basis set, and the PBE functional.

For bilayer MoTe_2 , the DFT calculations were performed using OpenMX with NC pseudopotentials, the **standard** multiple-zeta polarized basis set, and the PBE functional. SOC was neglected to keep the reference DFT calculations computationally tractable.

For a-Si, the training-set structures were generated via MD simulations using the Stillinger-Weber potential [101] implemented in the LAMMPS package. The simulation employed an NPT ensemble with a time step of 1 fs and followed a four-stage protocol: heating the system from 300 K to 2500 K over 30 ps, equilibrating at 2500 K for 1 ns, cooling down to 300 K over 2.2 ns, and finally equilibrating at 300 K for 150 ps. The subsequent DFT calculations were performed in Quantum ESPRESSO utilizing the PBE functional and optimized NC Vanderbilt pseudopotentials. The wavefunctions were expanded in a PW basis set with a kinetic energy cutoff of 60 Ry, and the Brillouin zone was sampled using a $3 \times 3 \times 3$ grid. To obtain the localized matrix representations, the converged PW Hamiltonians were subsequently projected onto the DZP basis set of SIESTA using DeepH-dock.

To generate the extensive structural dataset required for training the universal materials model, high-throughput OpenMX calculations managed by the AiiDA infrastructure were employed. These DFT calculations were performed with NC pseudopotentials, the **standard** multiple-zeta polarized basis set, and the PBE functional.

Data availability

The raw data supporting the findings in the “Results” section are available from <https://doi.org/10.5281/zenodo.19690642>, including the benchmark data sets, the trained neural network models, and predictions on the example systems.

Code availability

The first-principle calculation codes used for the current study, namely OpenMX, SIESTA, and Quantum

ESPRESSO, are all open-source software distributed via their corresponding websites. The DeepH-pack code introduced in this manuscript can be downloaded from <https://ticket.deeph-pack.com>, and <https://github.com/kYangLi/DeepH-pack-docs>. The DeepH-dock code can be downloaded from <https://github.com/kYangLi/DeepH-dock>.

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Author contributions

Y.L., Y.Z.W., and B.H.Z. contributed equally to this work. Y.L., W.H.D., and Y.X. coordinated code development. Y.L., Y.Z.W., B.H.Z., X.X.G., Y.X.W., Z.C.T., Z.X.W., Z.L.Y., J.L.L., M.H.S., Z.Z.C., H.G.T., B.C.W., Y.H.Y., and H.L. contributed to code development, dataset generation, and testing. F.H.J., W.H.D., and Y.X. supervised the project. All authors contributed to writing the manuscript.

Competing interests

Author W.H.D. is an Editorial Board Member of npj Computational Materials. W.H.D. was not involved in the journal's review of, or decisions related to, this manuscript. The other authors do not have a competing interest.

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Figure legends

Figure 1. Schematic of the DeepH computational workflow. Starting from the atomic structure, DeepH-pack predicts the material's density functional theory (DFT) Hamiltonian, enabling the prediction of various material properties with first-principles accuracy. DeepH-dock serves as the standardized data interface, converting data from various computational software packages (e.g., FHI-aims [42], Quantum ESPRESSO [44], SIESTA [45], ABACUS [46], OpenMX [47], *etc.*) for model training. DeepH-pack constitutes the core algorithmic component of the software, handling both model training and inference using existing models.

Figure 2. DeepH as a general-purpose neural network method. (a) Integration of DeepH with first-principles calculations. The DeepH model learns directly from converged density functional theory (DFT) data to predict electronic structures. These predictions serve a dual role: they enable efficient computation of material properties at first-principles accuracy, and provide high-quality initial guesses (such as charge density n , Hamiltonian H , and density matrix ρ) to accelerate DFT self-consistent field (SCF) convergence. (b) Physically constrained prediction of fundamental electronic structures. DeepH outputs scalar and tensor data that inherently satisfy physical constraints allowing systematic learning of diverse electronic properties. Its design ensures broad applicability across material systems and property types, establishing DeepH as a universal electronic structure algorithm.

Figure 3. DeepH workflow schematic. The DeepH workflow comprises three core stages: (a) Data preparation: User-defined atomic structures are processed using

density functional theory (DFT) software to compute electronic structures. DeepH modules then construct graph representations encoding atomic positions ($\{Z_i, \mathcal{R}_i\}$), overlap matrices (S), and Hamiltonian labels (H). (b) Model training: DeepH models are trained with graph data using suitable architectures and hyperparameters. The process minimizes loss functions to derive optimal parameters θ . (c) Prediction that includes model inference and Hamiltonian post-processing: Trained models predict Hamiltonians for new structures. DeepH-dock serves as an interface passing the Hamiltonian data to DFT software for continued calculations or post-processing tools for property analysis. (d) Internal mechanism of DeepH models: Atomic structures are encoded into SO(3)-equivariant neural representations. These are transformed into Hamiltonians via the inverse Clebsch-Gordan decomposition.

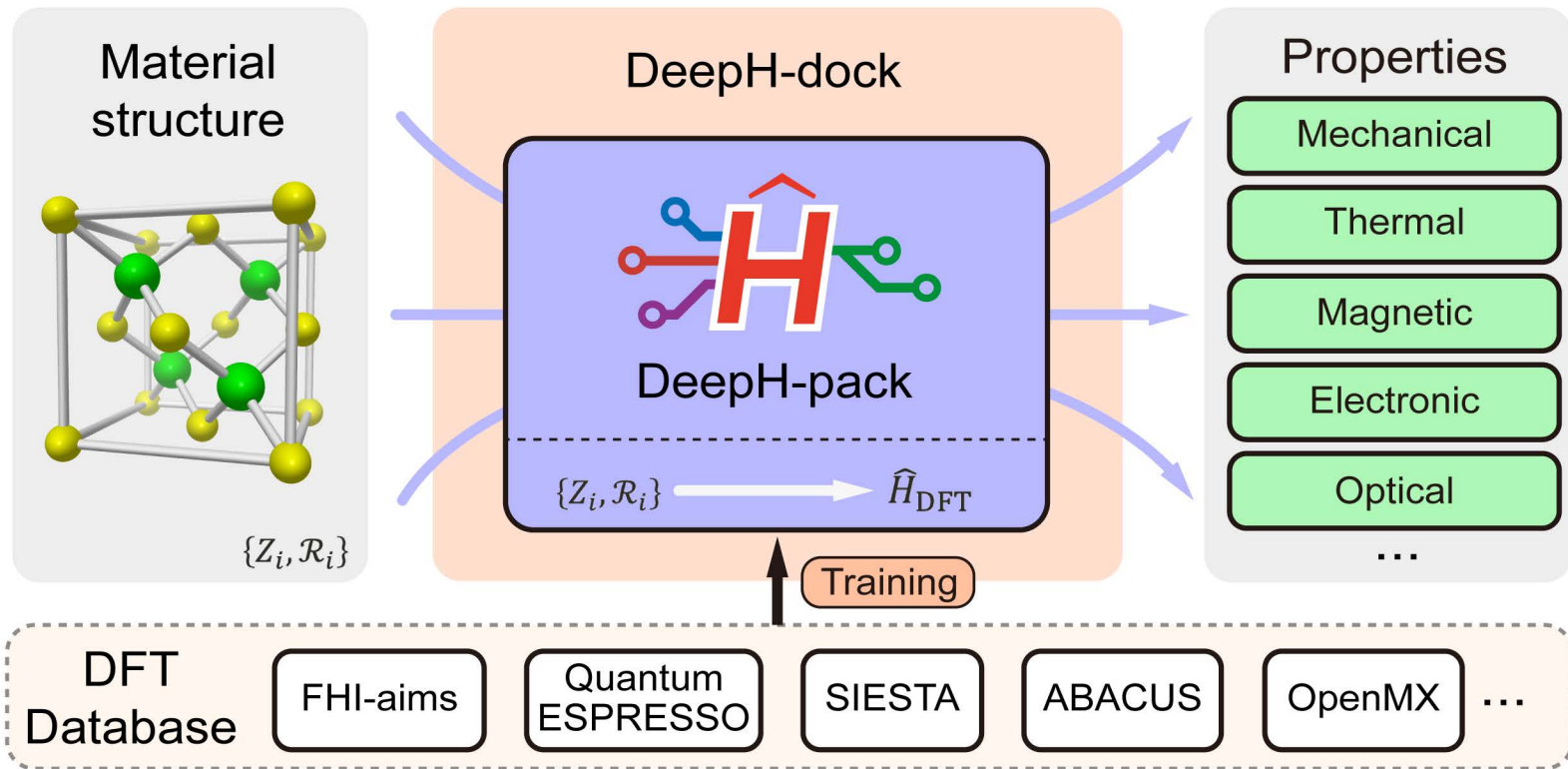
Figure 4. Schematic illustration of DeepH-dock functionality. (a) Serving as the interface for DeepH-pack, DeepH-dock converts data from multiple popular density functional theory (DFT) software packages into a unified format compatible with DeepH-pack. (b) DeepH-format data, either predicted by DeepH-pack or converted from other DFT software, can be further processed to compute target material properties. (c) DeepH-pack predicts Hamiltonians, enabling DeepH-dock to directly output key electronic properties (density of states, wavefunction visualization for the valence band maximum at Γ point, and band structure) for perturbed bilayer phosphorene.

Figure 5. Influence of training-set composition on DeepH’s prediction accuracy for a large-scale twisted bilayer MoTe_2 system. (a) Schematic diagram of the two training-set generation strategies. Strategy I includes interlayer-shifted bilayer and twisted bilayer structures. Strategy II adds a variety of uniquely designed quasi-1D moiré structures. (b) Band structure of a 1,986-atom twisted bilayer MoTe_2 predicted by the DeepH model trained using Strategy I (red solid lines), compared with the density functional theory (DFT) reference results (black dotted lines). The mean absolute error (MAE) of the band energies is 4.64 meV. (c) Band structure of the same system predicted by the model trained with Strategy II (red solid lines), showing improved agreement with the DFT reference (black dotted lines). The MAE is reduced to 1.26 meV.

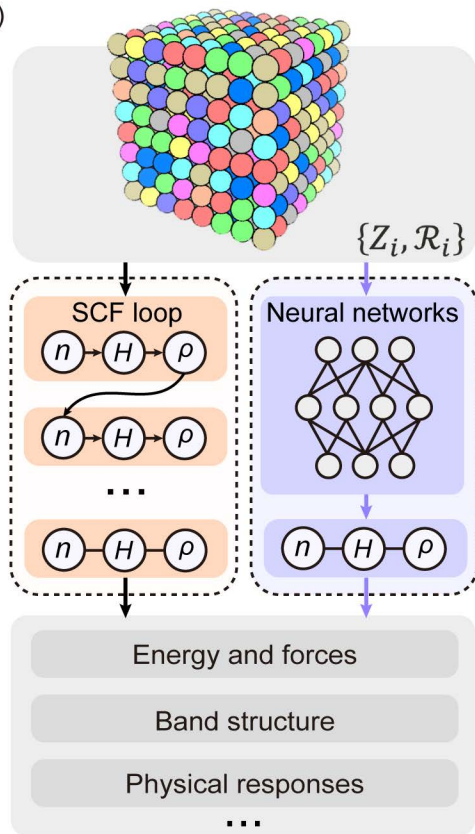
Figure 6. DeepH-based electronic structure prediction for large-scale amorphous silicon (a-Si). (a) Workflow for generating the training data. 300 independent a-Si configurations, each containing 216 atoms, were produced via melt-quench molecular dynamics using the Stillinger-Weber potential. The corresponding Hamiltonians calculated by density functional theory serve as training labels. (b) Density of states (DOS) predicted by DeepH for an a-Si system containing 4,096 atoms. (c) Effective band structure of the 4,096-atom a-Si system unfolded onto the Brillouin zone of FCC crystalline silicon.

The plot is visualized as a heat map of the local density of states in reciprocal space (qLDOS), representing the momentum-resolved spectral weight. The black dashed lines correspond to the band structure of single-crystal silicon.

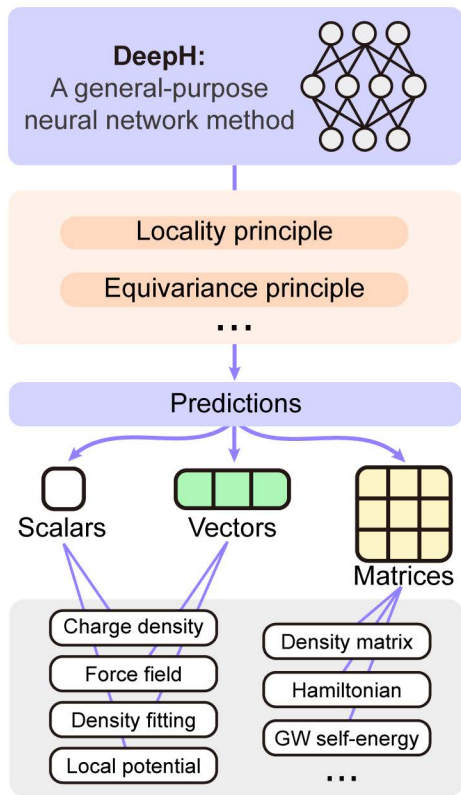
Figure 7. Prediction performance of the universal materials model of electronic structure. (a) The mean absolute errors (MAEs) of the predicted density functional theory (DFT) Hamiltonian are averaged for each element. The results reveal high prediction accuracy at the meV level. (b) Distribution of prediction errors across the entire test set, showing a mean error of 1.31 meV.



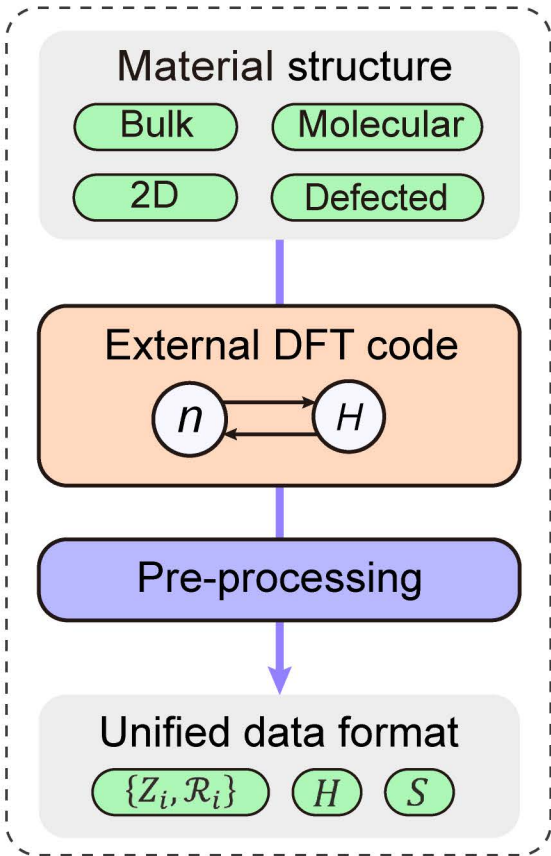
(a)



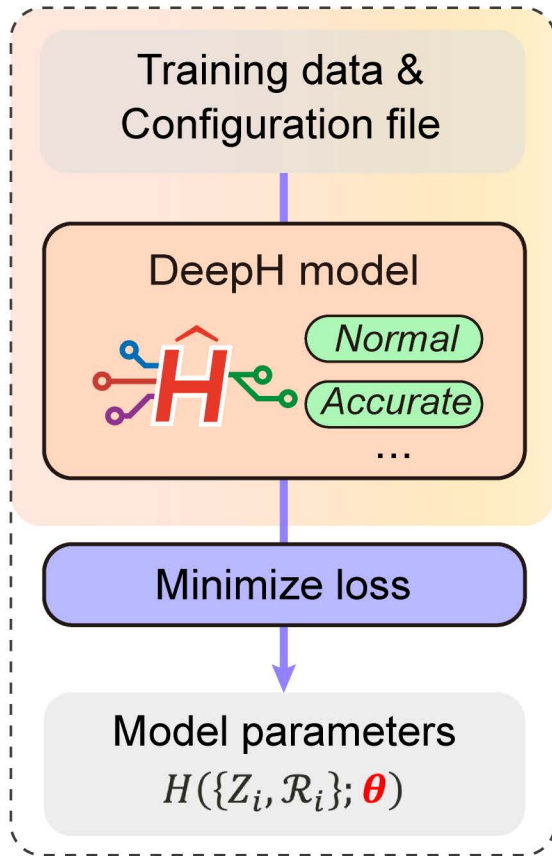
(b)



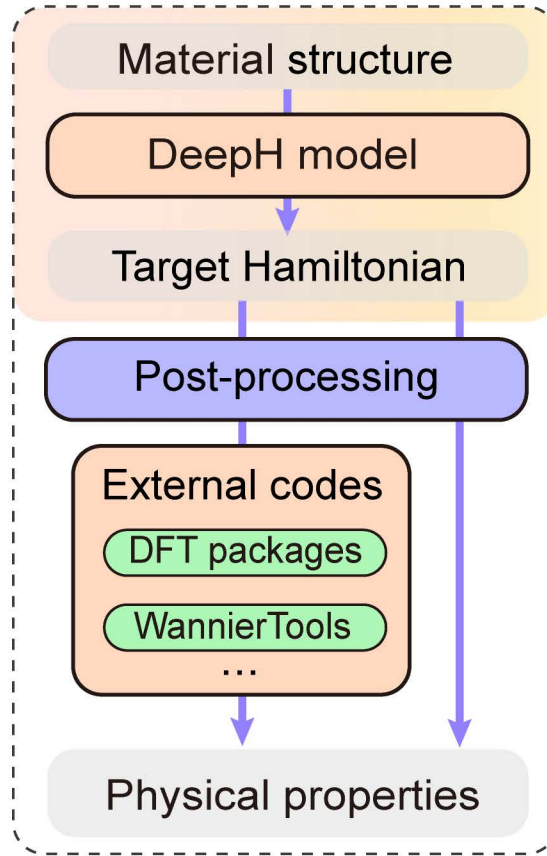
(a) Step 1: Data preparation



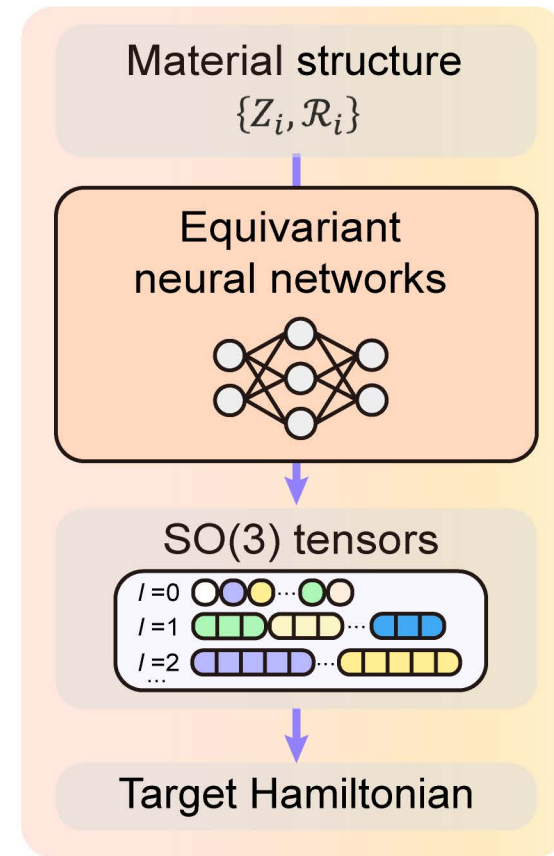
(b) Step 2: Model training

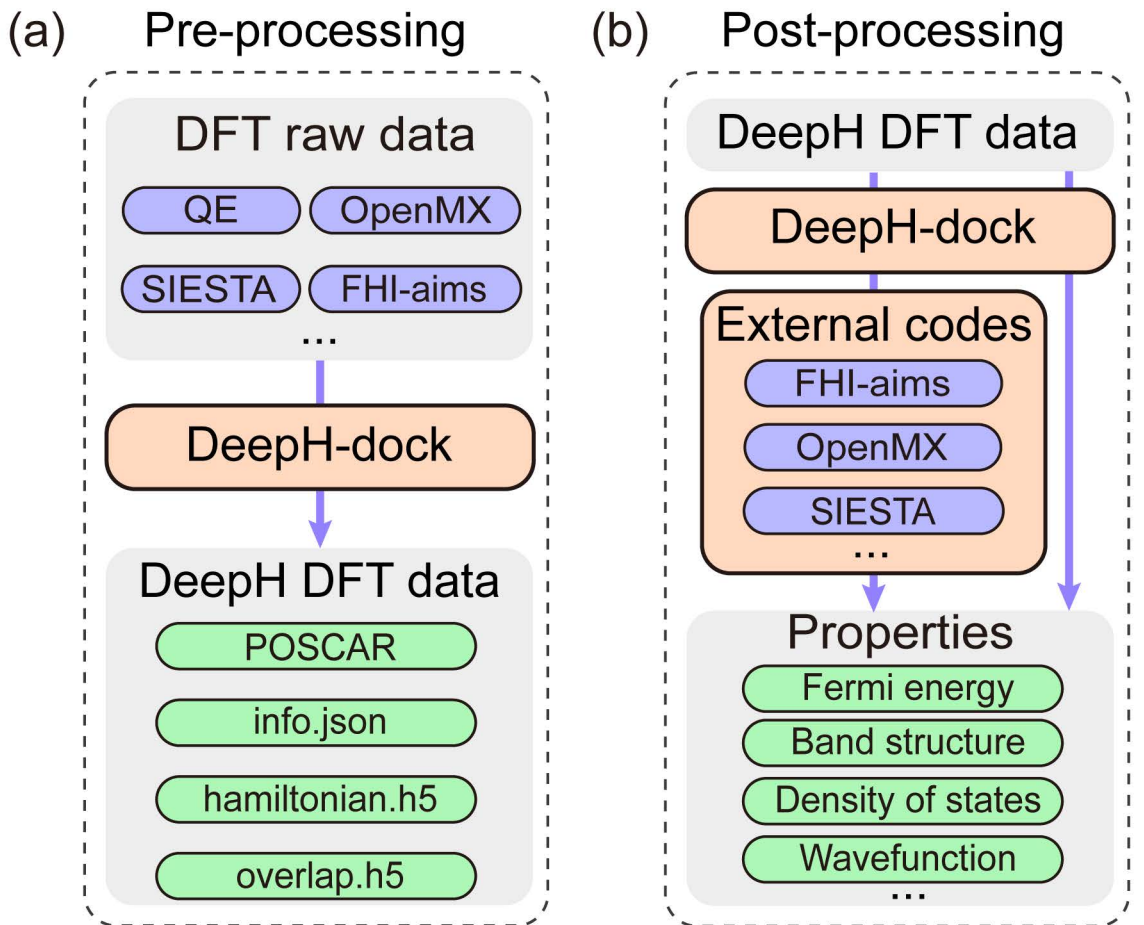


(c) Step 3: Prediction

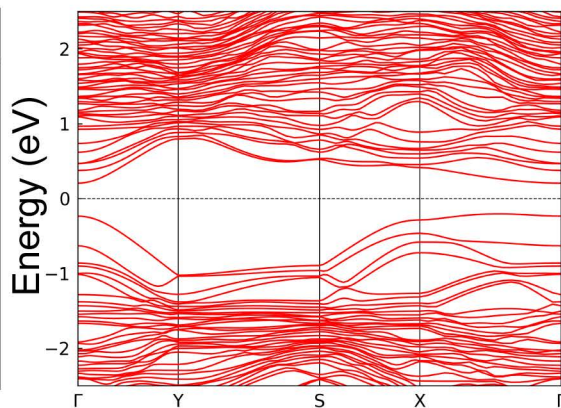
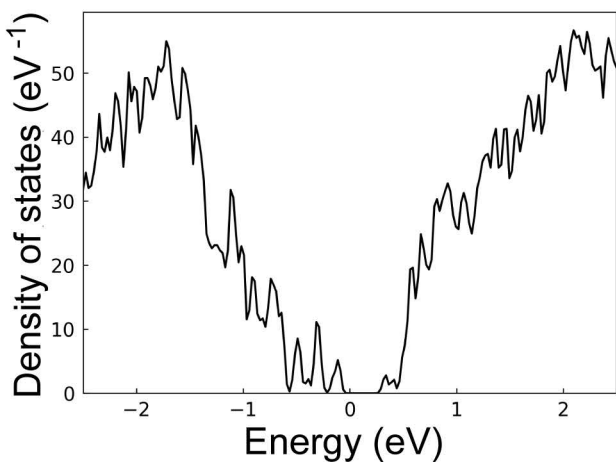
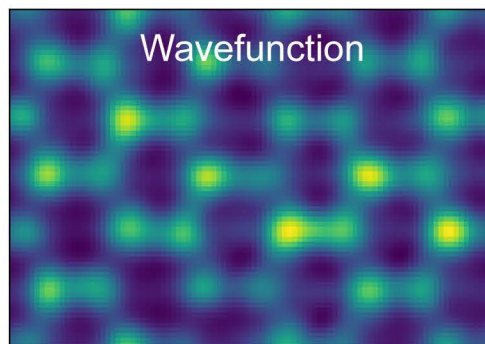
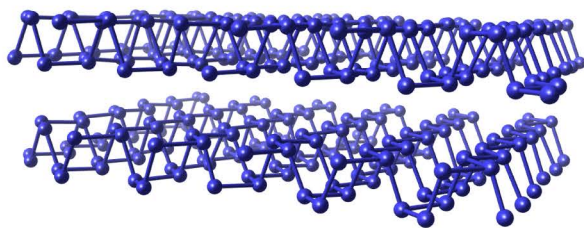


(d) DeepH architecture

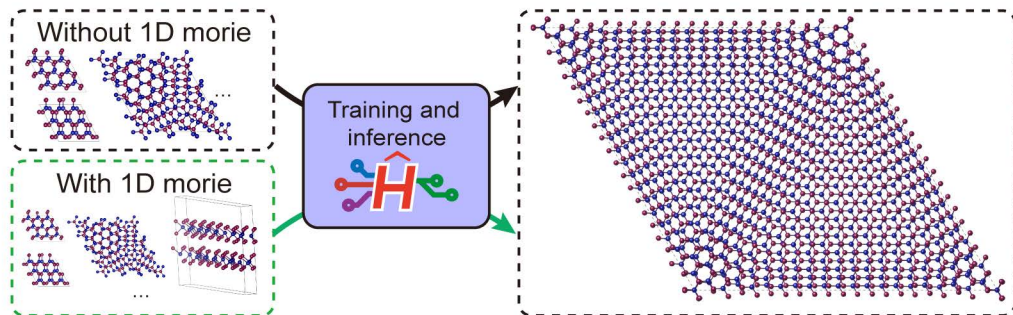




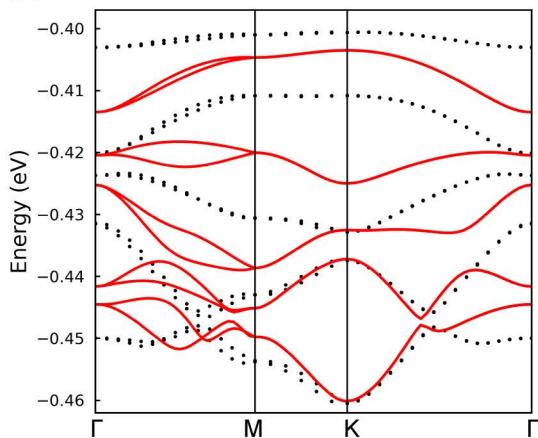
(c) Bilayer black phosphorus



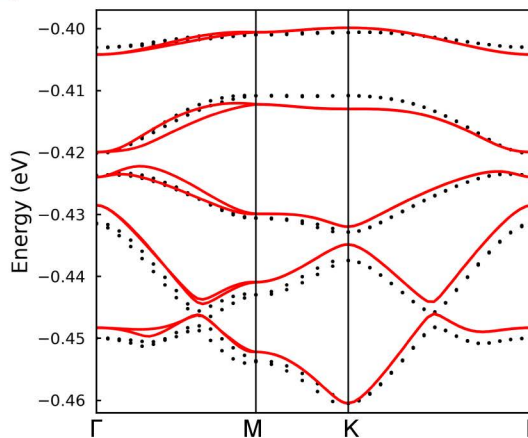
(a)



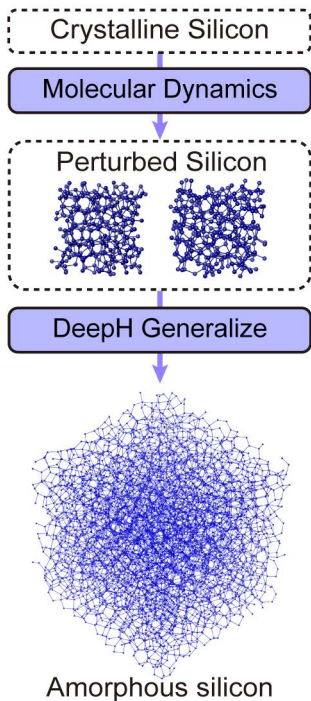
(b)



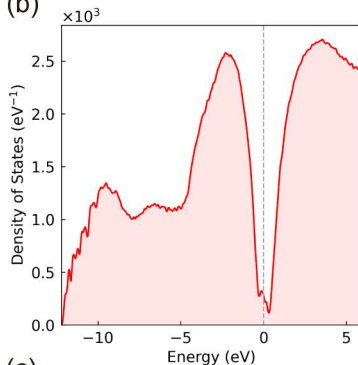
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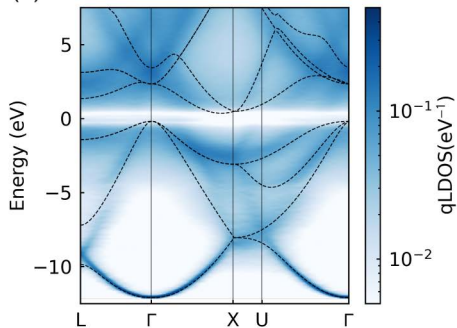
(a)



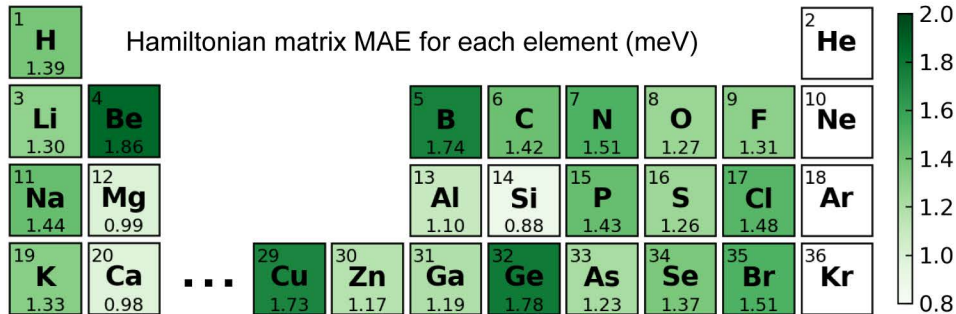
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(c)



(a)



(b)

